DATA VISUALIZATION · SUMMER 2026

Final Individual Project

Analyze, visualize, and tell a story with a real-world dataset of your choice

- 10+ analytical questions
- Publication-ready Plotly visuals
- Interactive Streamlit dashboard

Project Overview

Pick a **dataset that genuinely interests you**, develop meaningful analytical questions, and weave the answers into one compelling story — delivered three ways

Analysis notebook

Prelim Exploratory Data Analysis (EDA) + 10+ analytical questions, each answered with its own publication-ready visualization. **Built in Jupyter with Plotly**

Presentation

A slide deck (**exported to PDF**) summarizing your visuals, key insights, and conclusions — plus dashboard snapshots

Streamlit dashboard

A curated, interactive subset of your visuals, deployed live on **Streamlit Community Cloud** from a **GitHub repo**

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STEP 1

Choose a strong dataset

Your dataset MUST be…

- Real-world — sourced from a credible public repository/website, not artificially created for learning purposes
- Rich — enough columns to interrogate from many angles
- Varied — a mix of attribute types you can cross-reference

…and must NOT be

- Toy or artificially generated data (synthetic, random).
- Tired teaching sets — Iris, Titanic, Tips, mtcars, etc.
- Thin, single-type tables with little to explore.

Combine attribute types — the more, the richer your story: (text is also possible of course!)

Numerical

Categorical

Spatial

Temporal

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STEP 2

Pose 10+ genuinely analytical questions

An analytical question is multi-dimensional and complex enough to yield a useful insight — it relates variables, compares groups, or tracks change across time or space

Does NOT count

Preliminary, one-dimensional exploration:

- Value counts of a single column
- One-variable distributions / histograms
- Plain min / max / mean summaries
- "How many rows / categories are there?"

Counts as analytical

Multi-dimensional questions that drive insight:

- Relationships between two or more variables
- Comparisons across groups, regions or segments
- Trends and shifts over time or space
- Patterns conditioned on a third dimension

10 analytical questions = 10 visualizations, minimum

STEP 2 — EXAMPLES

From description to insight

Each weak prompt below only describes the data. Reworked, it forces a multi-dimensional answer worth visualizing

Preliminary — doesn't count		Analytical — what we want
How many Airbnb listings are in each London borough?	→	**How does price-per-guest vary across boroughs, and does it move with review scores?**
What is the distribution of CO_2 emissions?	→	**How has the link between GDP per capita and CO_2 per capita shifted by region since 1990?**
What are the most common energy sources?	→	**Which countries decoupled economic growth from fossil-fuel reliance over the last decade?**

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STEP 3

Build publication-ready visuals

Two non-negotiables

Plotly only — no Matplotlib, Seaborn, or any other plotting library

Explanatory, not exploratory — every figure delivers a concrete insight

Apply every design principle from the course

- Color Vision Deficiency (CVD) – safe & thoughtful palette throughout
- Declutter: remove gridlines, chart-junk and redundant ink
- Direct annotation on the figure as needed
- Titles that state the takeaway, not just the variables
- Muted grey for context + one highlight colour for focus
- Clean white background, clear visual hierarchy

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STEP 4

Design an interactive dashboard

Build a Streamlit app that presents a curated subset of your visuals and lets the reader explore — then deploy it live on Streamlit Community Cloud.

Curate, don't dump

- ✓ Show a thoughtful subset of your figures — not all ten
- ✓ Lead with the main highlights of your data (e.g. with important metrics)

Meaningful interactivity

- ✓ Filters and selectors that help the user to explore the data deeply
- ✓ Avoid dead-end dashboards that only restate one chart

Consistent design

- ✓ Carry the same CVD-safe palette
- ✓ Arrange the dashboard with clear hierarchy + decluttering across every view

Deploy it live

- ✓ Public GitHub repo + a working Streamlit Community Cloud URL

★ **Bonus:** a well-designed multi-tab dashboard earns extra credit. Use tabs to separate themes or audiences — but only if each tab is genuinely well structured

DELIVERABLES

What to hand in

Analysis notebook

- The Jupyter notebook (.ipynb) itself
- A PDF or HTML export of it
- The dataset file you used or a link to the csv online

Presentation

- Your visuals + key insights & conclusions
- Snapshots + link to the GitHub repo + URL of the live dashboard
- Exported to PDF

Streamlit dashboard

- Link to the GitHub source repo
- Link to the published Community Cloud dashboard
- (also embedded in the presentation)

Submit link to Public GitHub Final Project Repo (not your classwork one please!!!) · 1-to-1 message on Microsoft Teams · **No late submissions . Deadline Friday 31.07.2026**

ASSESSMENT

How it's graded

20%

Dataset selection

Real-world, rich, varied

40%

Methods / Ideas / Procedure

The analytical thinking — weighted highest

20%

Correctness

Sound, accurate analysis & code

20%

Quality of deliverables

Polish across notebook, deck & dashboard

The largest share of your grade rewards how you think — so invest in interesting, multi-dimensional questions

RESOURCES

Where to find a dataset

Strong starting points — and you may go beyond them.

1. **UCI Machine Learning Repository**
2. **Kaggle Datasets**
3. **Google Dataset Search**
4. **Amazon AWS Public Datasets**
5. **Data.gov**
6. **World Bank Open Data**
7. **FiveThirtyEight Datasets**
8. **European Data Portal**
9. **Information is Beautiful**

Tip: merging two or more sources to build your dataset is a helpful option in some topics

BEFORE YOU SUBMIT

Your final checklist

- A real-world dataset that's rich and mixes attribute types
- 10 multi-dimensional analytical questions = 10 explanatory visuals
- Plotly only — publication-ready, fully customized, CVD-safe
- An interactive Streamlit dashboard, deployed and linked
- All files zipped and sent on Teams — on time

One dataset. Ten questions. One story